Artem Semidetnov

Geneva, Switzerland artemsemidetnov@gmail.com artem.semidetnov.com

Education

MSc	**University of Geneva**, Mathematics (expected)	2025 – 2027
BSc	**Saint Petersburg State University**, Mathematics	2021 – 2025
BSc	**Neapolis University Pafos**, Applied Computer Science, JetBrains Program	2022 – 2024

Research Interests

- Categorical homotopy theory; simplicial methods and operads; homotopy type theory and computer-formalized approaches; geometric and combinatorial group theory.

Publications and Preprints

Operadic double dual (preprint in preparation)
Arshak Aivazian, Emmanuel Farjoun, Vasily Ionin, ***Artem Semidetnov***

An explicit non-elementary matrix over a bivariate Laurent polynomial ring Sept 2026
Vasily Ionin, ***Artem Semidetnov*** · arXiv:2609.04275

On Some More Problems from the Kourovka Notebook Aug 2026
Vasily Ionin, ***Artem Semidetnov*** · arXiv:2608.29219

The stable commutator length of a relator is not a one-relator group invariant Aug 2026
Artem Semidetnov · arXiv:2608.21465

The operad associated to a crossed simplicial group Dec 2025
Artem Semidetnov · arXiv:2512.14498

On the geometry of free nilpotent groups May 2021
Artem Semidetnov, Ruslan Magdiev · arXiv:2106.00095

Research Experience / Projects

On Simplicial Resolutions for Link Groups Oct 2026
Independent mathematical note presenting two proofs of the Lei-Li-Wu theorem on a simplicial resolution of link groups.

JetBrains - Research Intern, HoTT and Dependent Types Lab Pafos, Cyprus
June 2024 - Sept 2024

- Developed the official Arend library and formalized results in algebra and homotopy type theory.
- Formalized the Eckmann-Hilton argument, Eilenberg-Mac Lane spaces, homology of types, group automorphisms, Schur's lemma, Maschke's lemma, and characterizations of group actions; selected contributions shipped in Arend 1.10.

Selected Talks

The operad associated to a crossed simplicial group Copenhagen, Denmark
July 2026

- Young Topologists Meeting 2026

The operad associated to a crossed simplicial group Geneva, Switzerland
June 2026

- Groupes de Lie et espaces de modules seminar

Plus construction and simplicial methods — Pushkin, Russia, Aug 2025
- Conference "Algebraic Topology, Geometry, Combinatorics, and Data Analysis"

Twisting numbers on braid and Thompson groups — Nizhny Novgorod, Russia, 2024
- Topological Methods in Dynamics and Related Topics VII

On the Poisson boundary of lamplighter groups — Saint Petersburg, Russia
- St. Petersburg Seminar on Representation Theory and Dynamical Systems

On the geometry of free nilpotent groups — Novosibirsk, Russia, 2021
- Siberian Summer Conference

Honors and Awards

Excellence Master Fellowship, University of Geneva — 2025 – 2027

Rodnye Goroda Scholarship — 2023 – 2024

JetBrains Scholarship, Neapolis University Pafos — 2022 – 2024

Chebyshev Laboratory Travel Grant, Winter School in Mathematics and Theoretical Computer Science — 2025

Teaching Experience

Sirius Educational Center, Teaching Assistant, Groups of Intermediate Growth — Sochi, Russia, Apr 2024 – May 2024

Mathematics and Computer Science Faculty Program for Prospective Students, Teaching Assistant, Braid and Knot Theory — Saint Petersburg, Russia, June 2023

Laboratory for Continuous Mathematical Education, Mathematics Teacher for Gifted Students — Saint Petersburg, Russia, 2021 – 2023

Work Experience

JetBrains, Research Intern, HoTT and Dependent Types Lab — Remote, Nov 2025 – Apr 2026
- Designed and developed an MCP server for the Arend proof assistant, exposing type-checking and IDE functionality to AI coding agents.
- Integrated the MCP server with the Arend plugin for IntelliJ IDEA and built an experimental Arend coding agent using Koog.
- Fine-tuned models for Arend code completion.

IPONWEB (acquired by Criteo), Machine Learning Intern — Pafos, Cyprus, May 2023 – Sept 2023
- Reverse-engineered the behavior of a Criteo ML system that analyzes websites and produces word2vec representations; evaluated potential applications.

Workshops, Service and Additional Information

Teaching Higher Category Theory with Computers — ICERM, Brown University, Providence, RI, Aug 2026
Funded participant.

Academic Service and Outreach
- Organizer of the Euler International Mathematical Institute Functional Analysis Seminar.
- Invited judge, Saint Petersburg Tournament of Young Mathematicians (since 2021) and International Tournament of Young Mathematicians (2024, 2025).
- Problem author, ITYM 2024; Scientific Organizing Committee, ETEAM 2025.

Additional Information
Intel ISEF finalist (2020); English C1 (IELTS 8.0/9, taken in 2020 and 2024).